

Name: _____

Date: _____

Class: _____

Element C Initial Template

Design Requirement #1 (HIGH Priority)

The design must be able to organize and store up to 200 rolls of 3d printer filament.

One of the primary stakeholders, Mr. L, said that at the beginning of the year he restocks up to 200 rolls. This means that any storage system needs to be able to hold up to 200 rolls.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #2 (HIGH Priority)

Must be able to support up to 200 rolls weighing 1 kg each without breaking or sinking.

One of the primary stakeholders, Mr. L, stated that this system must store up to 200 rolls. This means that any storage needs to be capable of withstanding the up to 200kg of possible weight.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.

Design Requirement #3 (HIGH Priority)

Rolls must be organized by their specific color of filament, size of rolls, thickness of filament, and material of filament.

The primary stakeholder, Mr. L, stated that he needed a system that would organize the filaments to certain requirements such as specific color of filament, size of rolls, thickness of filament, and material of filament. More stakeholders, students of Mr. L, said that they have issues finding what filament they need when they need to print leading to them to ask Mr. L to get it for them.

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Design Requirement #4 (HIGH Priority)

The design needs to store 200 rolls in a compact manner, such that the ratio of vertical height of storage unit to horizontal length is greater than 1.

Mr. L has stated that he needs horizontal space especially on shelves in order to store his new 3D printers, and a compact storage unit would make organization easier while reducing clutter and messiness at the same time.

Share this design requirement with (a) primary stakeholder(s) to get their validation. Share any feedback, notes, and/or your revisions as evidence that you conducted primary stakeholder validation.